Relative Velocities

The relativity of velocities

No two objects can have a relative velocity greater than c. But what if a spacecraft traveling at 0.9c and it fires a missile which it observes to be moving at 0.8c with respect to it!? Velocities must transform according to the Lorentz transformation, and that leads to a rather non-intuitive result called Einstein velocity addition.

To analyse this situation we can consider two observers in relative motion with respect to each other who are both observing the motion of the missile. How do they measure the velocity of the object relative to each other if the speed of the object is close to that of light? We can consider a reference frame S' moving at a speed v relative to S. The missile has a velocity u' measured in the S' frame. The velocity u is the speed of the missile relative to frame S.

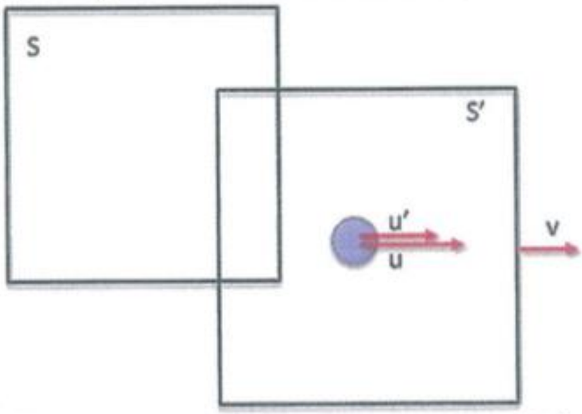


Figure 14.8: A reference frame S' is moving at a speed v relative to S. The missile shown by the blue circle has a velocity u' measured in the S' frame. The velocity u is the speed of the missile relative to frame S.

Then $u' = \dfrac{u - v}{1 - \dfrac{uv}{c^2}}$ and $u = \dfrac{u' + v}{1 - \dfrac{u'v}{c^2}}$

When v is much less than the speed of light c the denominator in the equation for u' approaches unity, and so $u' = u - v$, which is the Galilean velocity transformation equation. It is exactly what we would expect in the non-relativistic case. However, when u approaches the speed of light the equation becomes:

$$u' = \frac{c - v}{1 - \dfrac{cv}{c^2}} = \frac{c - v}{1 - \dfrac{v}{c}} = \frac{c\left(1 - \frac{v}{c}\right)}{1 - \dfrac{v}{c}} = c$$

From this result, we see that the speed of a particle travelling close to the speed of light measured by an observer in frame S is also measured as c by an observer in S'. This result is independent of the relative motion of S and S'. This s consistent with Einstein's second postulate, that the speed of light must be c relative to all inertial reference frames. Furthermore, we find that the speed of an object can never be measured to be greater than c.

Relativistic Addition of Velocities

No two objects can have a relative velocity greater than c.

e.g. A spacecraft moving at 0.9c fires a missile moving at 0.8c relative to it.

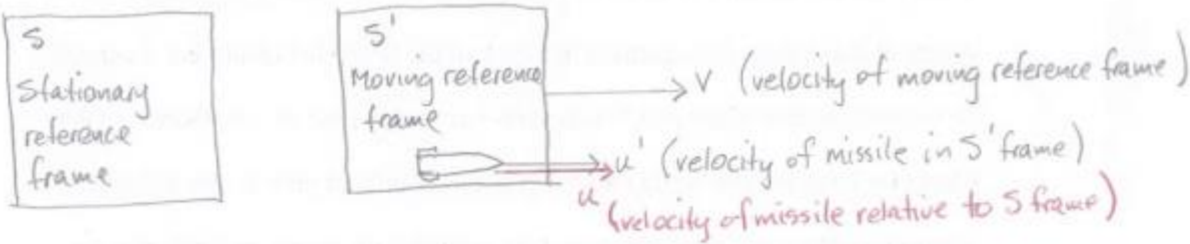
A stationary observer sees this as 1.7c, which is impossible.

Velocities must transform according to the Lorentz transformation, leading to a non-intuitive result called Einstein velocity addition.

(See Ch 8.4 - Relativistic addition of velocities)

Solution to the "problem".

Consider two observers in relative motion to each other and both are observing the motion of the missile.



The equation describing u is given by:

$$\boxed{u = \frac{u' + v}{1 + \frac{u'v}{c^2}}}$$

where u = velocity of the moving object in the stationary frame.
u' = velocity of the moving object in the moving frame. $(= 0.8c)$
v = velocity of the moving frame of reference $(= 0.9c)$
c = speed of light.

ie. $$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

$$= \frac{0{\cdot}8c + 0{\cdot}9c}{1 + \frac{(0{\cdot}8c)(0{\cdot}9c)}{c^2}}$$

$$= \frac{1{\cdot}7c}{1{\cdot}72}$$

$$= \underline{0{\cdot}988c}$$

EXAMPLE

Two spaceships are moving apart from each other, A at 0·7c and B at 0·5c in the other direction. What is the velocity of A as seen by an observer in B?

Consider B as the stationary reference frame.

Take the direction of B as +ve.

$V = -0{\cdot}7c$

$u = 0{\cdot}5c$

u' = velocity of A relative to B

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}}$$

$$= \frac{0{\cdot}5c - (-0{\cdot}7c)}{1 - \frac{(0{\cdot}5c)(-0{\cdot}7c)}{c^2}}$$

$$= \frac{1{\cdot}2c}{1{\cdot}35}$$

$$= 0{\cdot}889c$$

∴ Spaceship A is receding at 0·889c from B.